WORKED SOLUTION

STEP II 2005, Question 8

Pure • Separable Differential Equations

STEP II 2005, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q8

The key idea

The equation is separable: all the y-dependence is a single factor y^2, so dividing by it gives $\frac{1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x}$, the derivative of $-1/y$. The x-integral yields to the substitution $u = 1 + x^2$, because the odd power x^3 supplies the $x\,\mathrm{d}x$ that u needs. For the second curve, notice that the right-hand side is z^3 rather than y^2: the quantity to look at is then $1/z^2$, and that turns out to obey exactly the same equation as $1/y$.

Solution

Solving for y

On C we have $y(0) = 1 > 0$, and the right-hand side of the equation is never negative for $x \geq 0$, so y never decreases. Hence $y \geq 1$ throughout, and we may divide by y^2:

$$\frac{1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^3}{(1+x^2)^{5/2}} \quad\Longrightarrow\quad -\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{y}\right) = \frac{x^3}{(1+x^2)^{5/2}}.$$

For the integral put $u = 1 + x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$ and $x^3\,\mathrm{d}x = x^2 \cdot x\,\mathrm{d}x = \frac{1}{2}(u-1)\,\mathrm{d}u$:

$$\begin{aligned}\int \frac{x^3}{(1+x^2)^{5/2}}\,\mathrm{d}x &= \frac{1}{2}\int (u^{-3/2} - u^{-5/2})\,\mathrm{d}u = -u^{-1/2} + \tfrac{1}{3}u^{-3/2} + \text{const}\\ &= -\frac{3u-1}{3u^{3/2}} + \text{const} = -\frac{2+3x^2}{3(1+x^2)^{3/2}} + \text{const}.\end{aligned}$$

Therefore

$$\frac{1}{y} = \frac{2+3x^2}{3(1+x^2)^{3/2}} + k$$

for some constant k. At $x = 0$ this reads $1 = \frac{2}{3} + k$, so $k = \frac{1}{3}$, which is the required result:

$$\frac{1}{y} = \frac{2+3x^2}{3(1+x^2)^{3/2}} + \frac{1}{3}.$$

Large x

Take out the dominant powers, $3x^2$ from the numerator and x^3 from $(1+x^2)^{3/2}$:

$$\frac{2+3x^2}{3(1+x^2)^{3/2}} = \frac{3x^2(1+\frac{2}{3x^2})}{3x^3(1+\frac{1}{x^2})^{3/2}} = \frac{1}{x}\Big(1+\frac{2}{3x^2}\Big)\Big(1-\frac{3}{2x^2}+\cdots\Big) = \frac{1}{x}+O\Big(\frac{1}{x^3}\Big),$$

using the binomial expansion, which is valid since $1/x^2 < 1$ for large x. So

$$\frac{1}{y} = \frac{1}{3}+\frac{1}{x}+O\Big(\frac{1}{x^3}\Big) = \frac{1}{3}\Big(1+\frac{3}{x}+O\Big(\frac{1}{x^3}\Big)\Big),$$

and inverting, with $(1+t)^{-1} = 1-t+O(t^2)$,

$$y = 3\Big(1-\frac{3}{x}+O\Big(\frac{1}{x^2}\Big)\Big) \approx 3-\frac{9}{x},$$

as required.

Ask yourself Why can we throw away the $\frac{2}{3x^2}$ and $-\frac{3}{2x^2}$ terms but not the $\frac{1}{x}$?

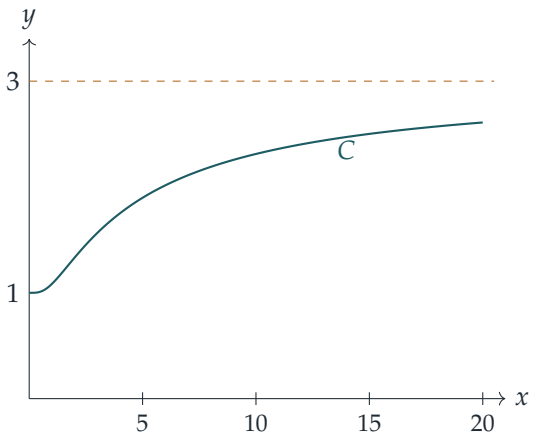
The target $3-\frac{9}{x}$ is accurate to order $\frac{1}{x}$. In $1/y$ the correction terms are a further factor $\frac{1}{x^2}$ smaller than $\frac{1}{x}$, so they only affect y at order $\frac{1}{x^2}$ or beyond. The $\frac{1}{x}$ term itself is what produces the $-\frac{9}{x}$.

Sketch of C

We now collect the features of the curve.

- $y(0) = 1$, and $\mathrm{d}y/\mathrm{d}x = 0$ at $x = 0$. Near the origin $\mathrm{d}y/\mathrm{d}x \approx x^3$, so $y \approx 1+\frac{1}{4}x^4$: the curve starts out very flat.
- For $x > 0$ the gradient is strictly positive, so y is increasing.
- The first term of $1/y$ is positive, so $1/y > \frac{1}{3}$, giving $1 \le y < 3$ for all x. The curve never blows up.
- As $x \to \infty$, $y \to 3$ from below, like $3-\frac{9}{x}$. The approach is slow: $y(10) \approx 2.31$.

Since the curve starts out concave up and then levels off towards $y = 3$, it must have a point of inflection.



The curves for z

Wherever $z \neq 0$ we can divide by z^3. Since $\dfrac{\mathrm{d}}{\mathrm{d}x}(z^{-2}) = -2z^{-3}\dfrac{\mathrm{d}z}{\mathrm{d}x}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{z^2}\right) = -\frac{2}{z^3}\cdot\frac{x^3z^3}{2(1+x^2)^{5/2}} = -\frac{x^3}{(1+x^2)^{5/2}},$$

which is exactly the equation satisfied by $1/y$. On both curves $1/z^2 = 1$ at $x = 0$, the same value as $1/y$. So

$$\frac{1}{z^2} = \frac{1}{y}, \qquad \text{that is} \qquad z^2 = y.$$

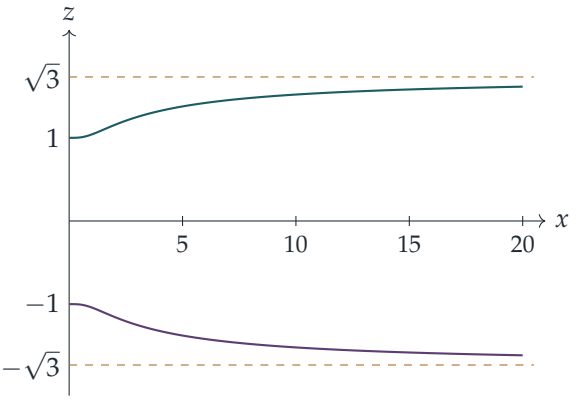
This also confirms that z never reaches 0, which justifies the division. While $z \neq 0$ we have $z^2 = y \geq 1$, so z can't get close to 0. By continuity, z therefore keeps the sign it starts with, and the two curves are

$$z = \sqrt{y} \quad (z(0) = 1), \qquad z = -\sqrt{y} \quad (z(0) = -1).$$

The upper curve rises from 1 towards $\sqrt{3}$, again starting flat (now $z \approx 1 + \frac{1}{8}x^4$), and for large x

$$z = \sqrt{3}\left(1 - \frac{3}{x} + \cdots\right)^{1/2} \approx \sqrt{3} - \frac{3\sqrt{3}}{2x}.$$

The lower curve is its reflection in the x-axis: it falls from -1 towards the asymptote $z = -\sqrt{3}$.



Ask yourself Why isn't the curve with $z(0) = -1$ just a copy of the other one, translated or otherwise transformed?

The equation is odd in z: replacing z by $-z$ changes both sides by a factor -1. So if $z(x)$ is a solution then so is $-z(x)$, and the solution starting at -1 is the reflection of the one starting at $+1$. This matches $z = \pm\sqrt{y}$.

Checks

- Differentiate the answer: with $1/y = u^{-1/2} - \frac{1}{3}u^{-3/2} + \frac{1}{3}$, we get $\mathrm{d}(1/y)/\mathrm{d}x = 2x\left(-\frac{1}{2}u^{-3/2} + \frac{1}{2}u^{-5/2}\right) = -x^3u^{-5/2}$, as needed.
- A numerical (Runge–Kutta) solution of the original equation gives $y(1) = 1.08391$ and $y(5) = 1.89777$, agreeing with the formula to 10 or more significant figures. Solving the z-equation numerically gives $z(5) = \pm 1.37759 = \pm\sqrt{y(5)}$.
- At $x = 100$ the formula gives $y = 2.9126$, against $3 - \frac{9}{x} = 2.91$. The difference is about $27/x^2$, consistent with an $O(1/x^2)$ error.

Common slips

- Losing the minus sign: $\int y^{-2}\,\mathrm{d}y = -y^{-1}$.
- Concluding that $y \to 3$ quickly because $3 - \frac{9}{x}$ "looks close" to 3. At $x = 10$ the curve is still only at about 2.3.
- Writing $z = \sqrt{y}$ for both curves, or failing to explain why z cannot change sign.

Beyond STEP

Both equations have the form $\mathrm{d}y/\mathrm{d}x = f(x)\,y^n$, a special case of a *Bernoulli equation*, which the substitution $w = y^{1-n}$ always turns into a linear equation. We used exactly this substitution: $w = 1/y$ and $w = 1/z^2$. The step "same equation, same initial value,

so same function" is an instance of the existence and uniqueness theorem (Picard–Lindelöf) for first-order equations. Boyce and DiPrima, *Elementary Differential Equations*, cover both.

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